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Video Prediction is challenging.

Decomposition

- | | | | | | | | | | | |
|----------------|--|--|--|--|--|--|--|--|--|--|
| Groundtruth | | | | | | | | | | |
| Blurry Results | | | | | | | | | | |

Ground truth

3x5

3x3 3x5

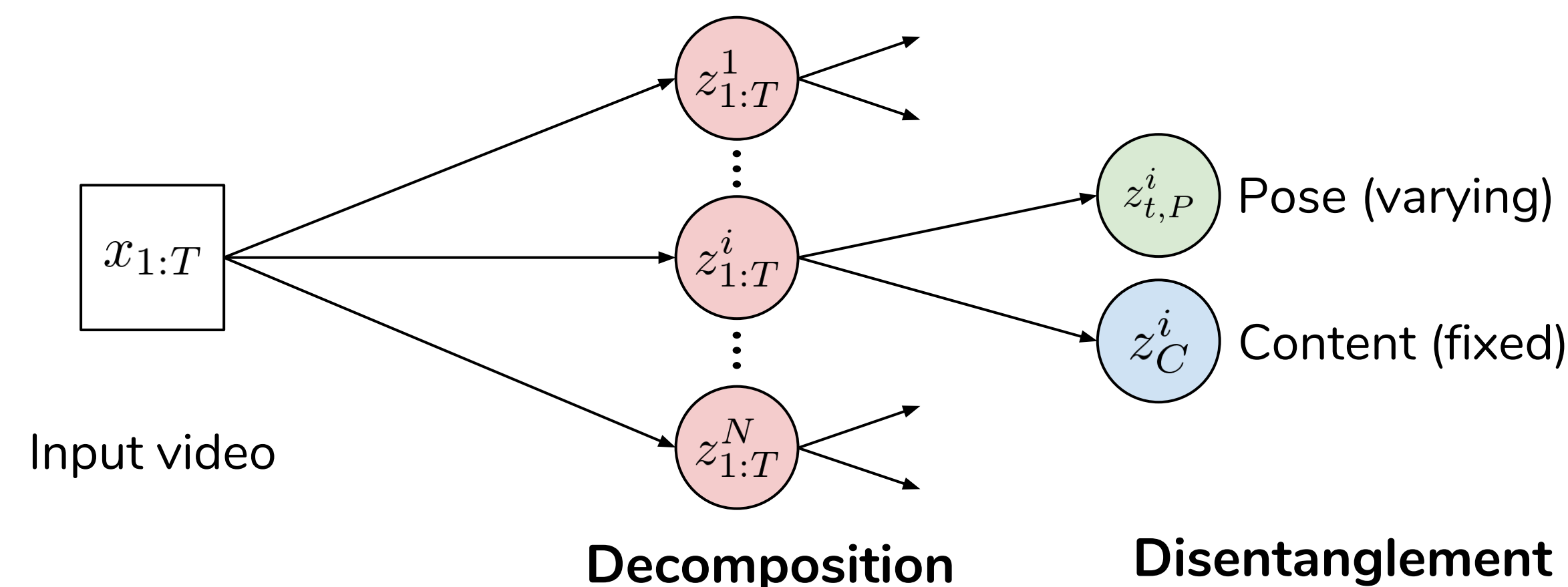
3x5 3x3

Disentanglement

- Content (fixed)
- Pose (varying)          

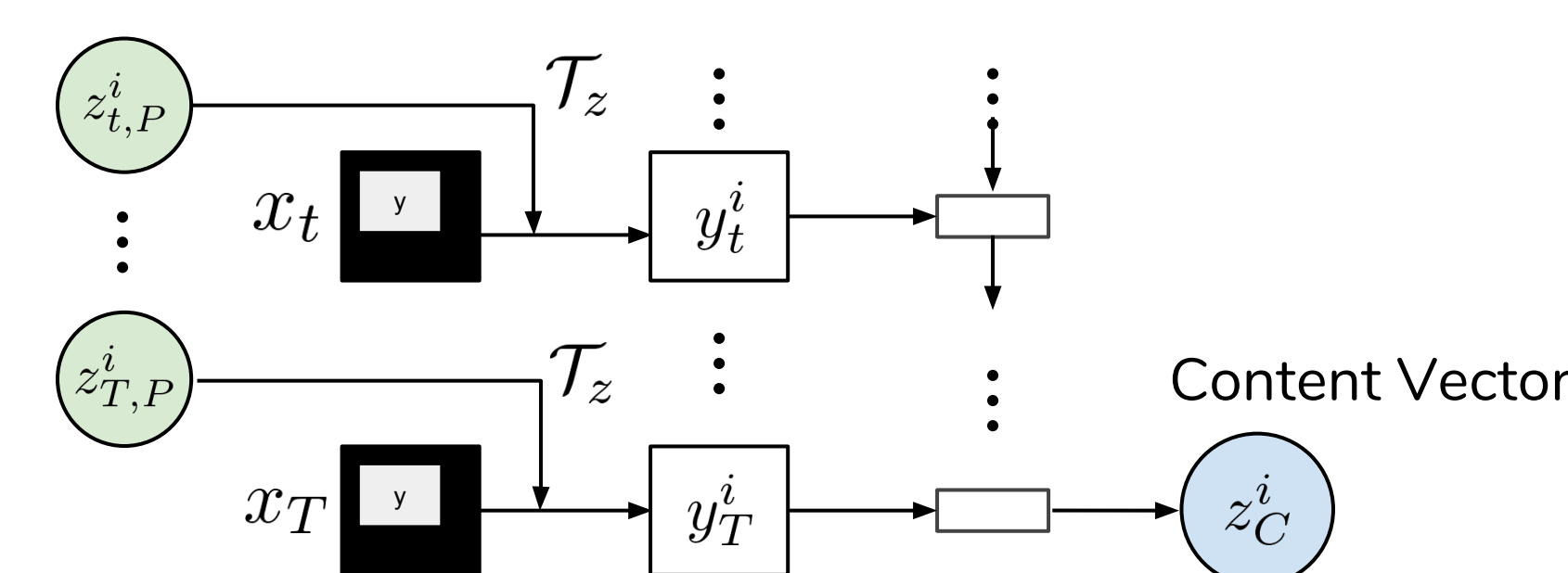
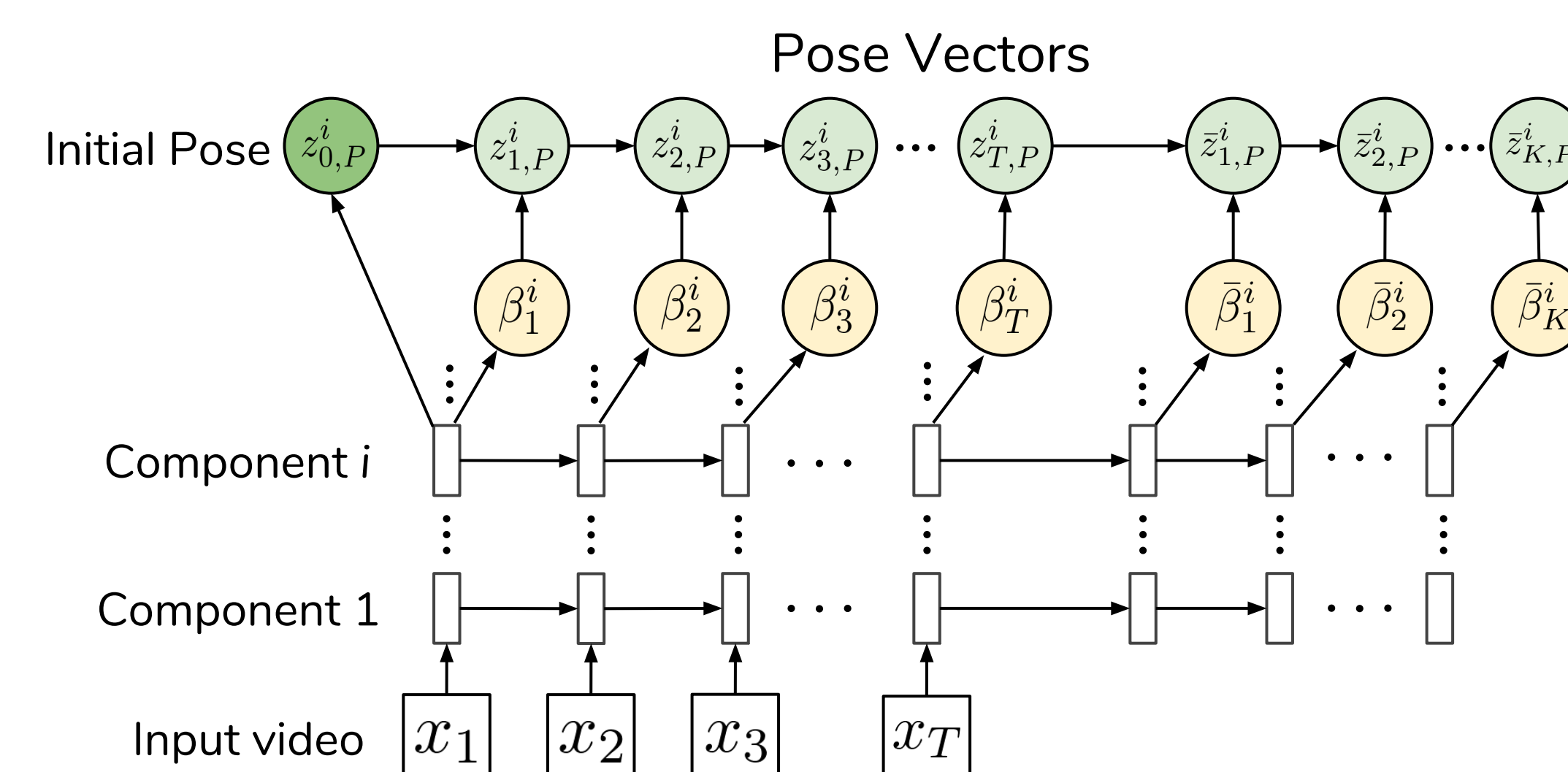
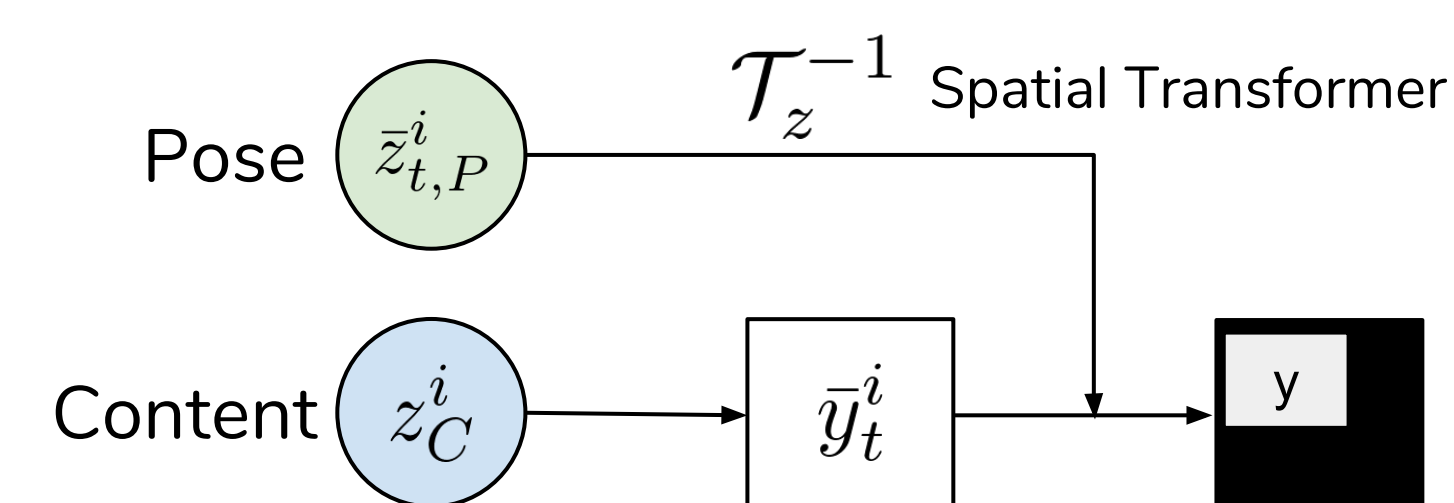
Overview: Given an input video, decompose and disentangle the random latent variables.

- **Decompose** into N components.
- **Disentangle** each component into one fixed content vector and time-varying pose vectors.



Frame Generation $p(\bar{x}_{1:K}^i \mid \bar{z}_{1:K}^i)$
from pose and content.

- Generate rectified image from the content vector with deconvolution layers.
- Use spatial transformer (parametrized by pose) to warp the image onto the frame.



Pose vectors: Time-varying pose vectors.

- Use 2D recurrence, one for temporal modeling and one for decomposition.
- Generate an initial pose and transition variables for each component.
- Combine to get pose vectors.

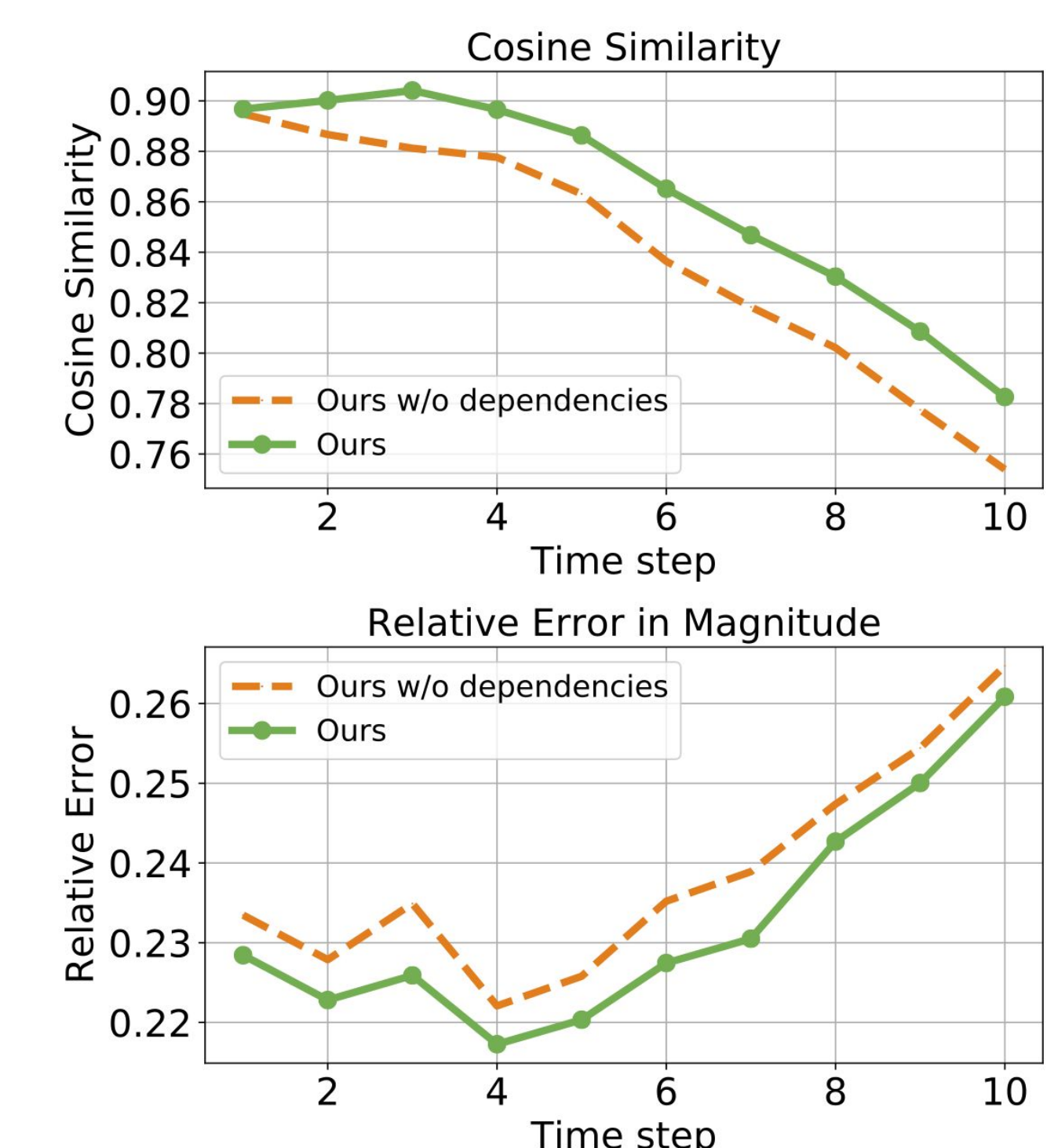
Content vector: Single fixed content vector.

- Use spatial transformer to extract a rectified image for every pose vector.
- Use another RNN as pooling to obtain a single content vector.

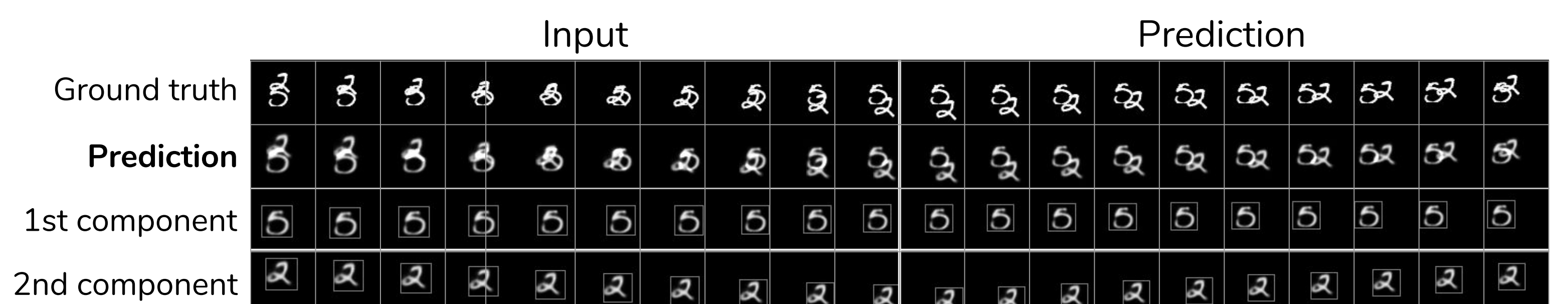
Moving MNIST results

Model	BCE	MSE
Shi et al. [45]	367.2	-
Srivastava et al. [33]	341.2	-
Brabandere et al. [5]	285.2	-
Patraucean et al. [26]	262.6	-
Ghosh et al. [10]	241.8	167.9
Kalchbrenner et al. [15]	87.6	-
MCNet [39]	1308.2	173.2
DRNet [6]	862.7	163.9
Ours w/o Decomposition	325.5	77.6
Ours w/o Disentanglement	296.1	<u>65.6</u>
Ours (DDPAE)	<u>223.0</u>	38.9

Bouncing Balls results



Moving MNIST



Bouncing Balls

